

Advanced Mathematical Techniques

Bookwork 7

1. **Problem 5.9.8:** Prove that:

$$\begin{aligned} \int_0^\infty \frac{x^n e^x dx}{(e^x - 1)^2} &= n! \zeta(n) \\ \int_0^\infty \frac{x^{n-1}}{e^{ax} - 1} dx &= \frac{1}{a^n} \Gamma(n) \zeta(n) \\ \frac{\partial}{\partial a} \int_0^\infty \frac{x^{n-1}}{e^{ax} - 1} &= - \int_0^\infty \frac{x^n e^{ax}}{(e^{ax} - 1)^2} dx = -a^{-(n+1)} \Gamma(n+1) \zeta(n) \\ \int_0^\infty \frac{x^n e^{ax}}{(e^{ax} - 1)^2} dx &= a^{-(n+1)} \Gamma(n+1) \zeta(n) \\ a = 1, \int_0^\infty \frac{x^n e^x dx}{(e^x - 1)^2} &= \Gamma(n+1) \zeta(n) = n! \zeta(n) \quad \checkmark \end{aligned}$$

2. **Problem 5.9.9:** The Bloch-Gruneisen approximation for the resistance in a monovalent metal is:

$$\rho = C \frac{T^5}{\Theta^6} \int_0^{\Theta/T} \frac{x^5 dx}{(e^x - 1)(1 - e^{-x})}$$

where Θ is the Debye temperature characteristic of the metal.

(a) For $T \rightarrow \infty$, show that

$$\rho \approx \frac{C}{4} \cdot \frac{T}{\Theta^2}$$

$$\begin{aligned} \int_0^{\Theta/T} \frac{x^5 dx}{(e^x - 1)(1 - e^{-x})} &= \int_0^{\Theta/T} \frac{x^5 e^x dx}{(e^x - 1)^2} \xrightarrow{\Theta/T \approx 0, x \approx 0} \int_0^{\Theta/T} \frac{x^5 dx}{(1 + x - 1)^2} = \int_0^{\Theta/T} x^3 dx = \frac{1}{4} \left(\frac{\Theta}{T} \right)^4 \\ C \frac{T^5}{\Theta^6} \cdot \frac{1}{4} \left(\frac{\Theta}{T} \right)^4 &= \frac{C}{4} \cdot \frac{T}{\Theta^2} \quad \checkmark \end{aligned}$$

(b) For $T \rightarrow 0$, show that

$$\rho \approx 5! \zeta(5) C \frac{T^5}{\Theta^6}$$

$$\begin{aligned} T \rightarrow 0, \Theta/T \rightarrow \infty, \int_0^\infty \frac{x^5 dx}{(e^x - 1)(1 - e^{-x})} &= 5! \zeta(5) \quad (5.9.8) \\ \therefore \rho &\approx 5! \zeta(5) C \frac{T^5}{\Theta^6} \quad \checkmark \end{aligned}$$

3. **Problem 8.2.14:** The total energy radiated by a blackbody is given by:

$$u = \frac{8\pi k^4 T^4}{c^3 h^3} \int_0^\infty \frac{x^3}{e^x - 1} dx$$

Show that the integral portion is equal to $3! \zeta(4) = 6\zeta(4)$.

$$\begin{aligned} \int_0^\infty \frac{x^3}{e^x - 1} dx &= \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx = \int_0^\infty \sum_{k=1}^\infty x^3 e^{-kx} dx = \sum_{k=1}^\infty \int_0^\infty x^3 e^{-kx} dx \\ u = mx, dx &= \frac{du}{m}, \text{ Above} = \sum_{k=1}^\infty \frac{1}{k^4} \int_0^\infty u^3 e^{-u} du = \Gamma(4) \zeta(4) = 3! \zeta(4) \quad \checkmark \end{aligned}$$